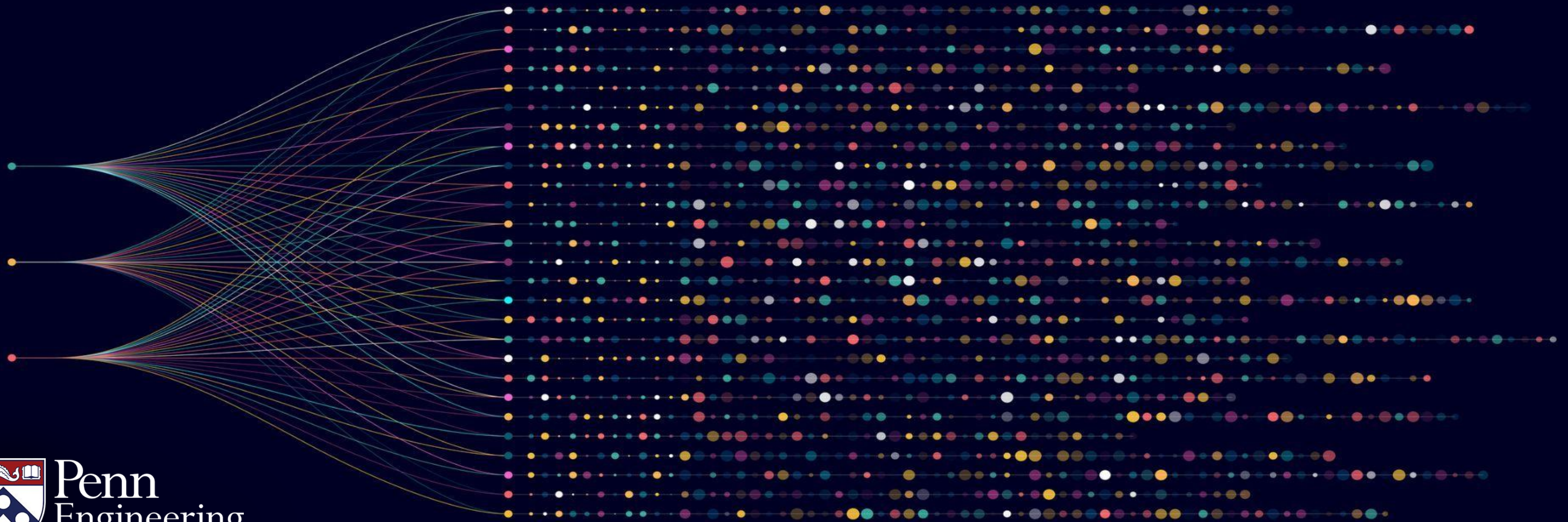


Algorithms for Big Data

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The power of inclusion - exclusion



Inclusion exclusion principle

Elementary tool from combinatorics:

Inclusion-exclusion principle – used to reason about the size of the union of sets

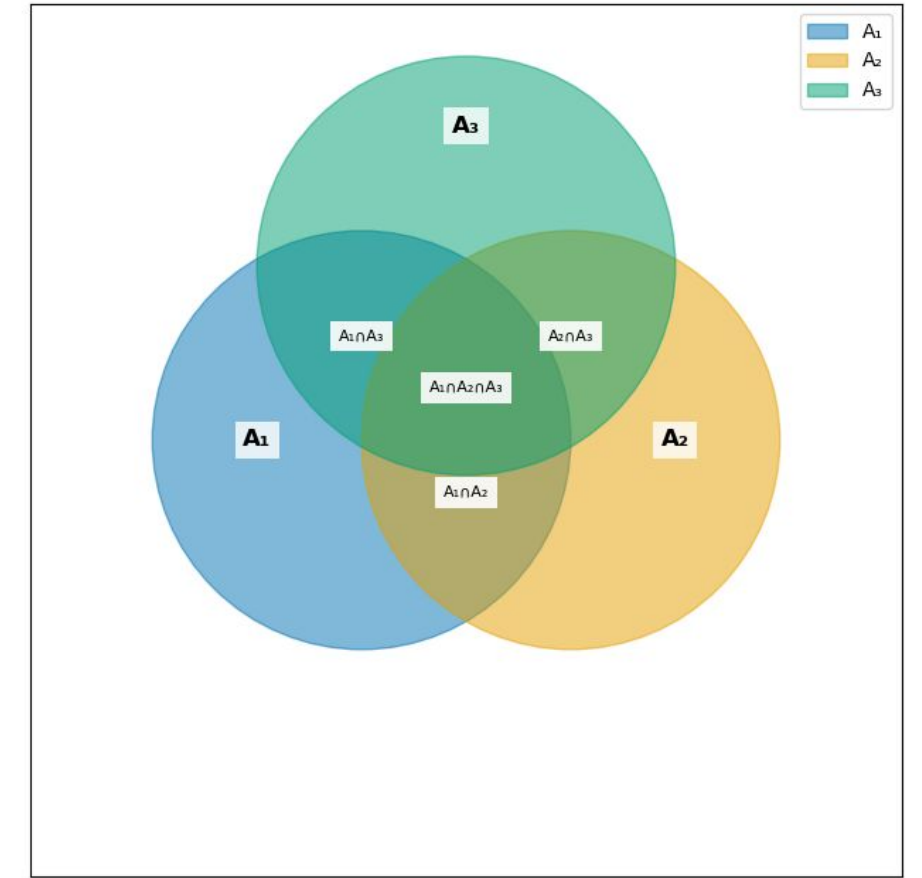
Roughly speaking: the inclusion principle gives a series where we alternately overestimate and underestimate the size of the union

Inclusion Exclusion principle

Suppose we have three finite sets A_1, A_2, A_3

Inclusion exclusion principle says that

$$|A_1 \cup A_2 \cup A_3| \leq |A_1| + |A_2| + |A_3| - (1).$$



$$|A_1 \cup A_2 \cup A_3| \geq |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| - (2).$$

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3| - (3).$$

Equality in (3) because the number of sets here is 3. Inclusion exclusion can be extended to arbitrary number of sets

Inclusion Exclusion principle

Suppose we have three finite sets A_1, A_2, A_3 .

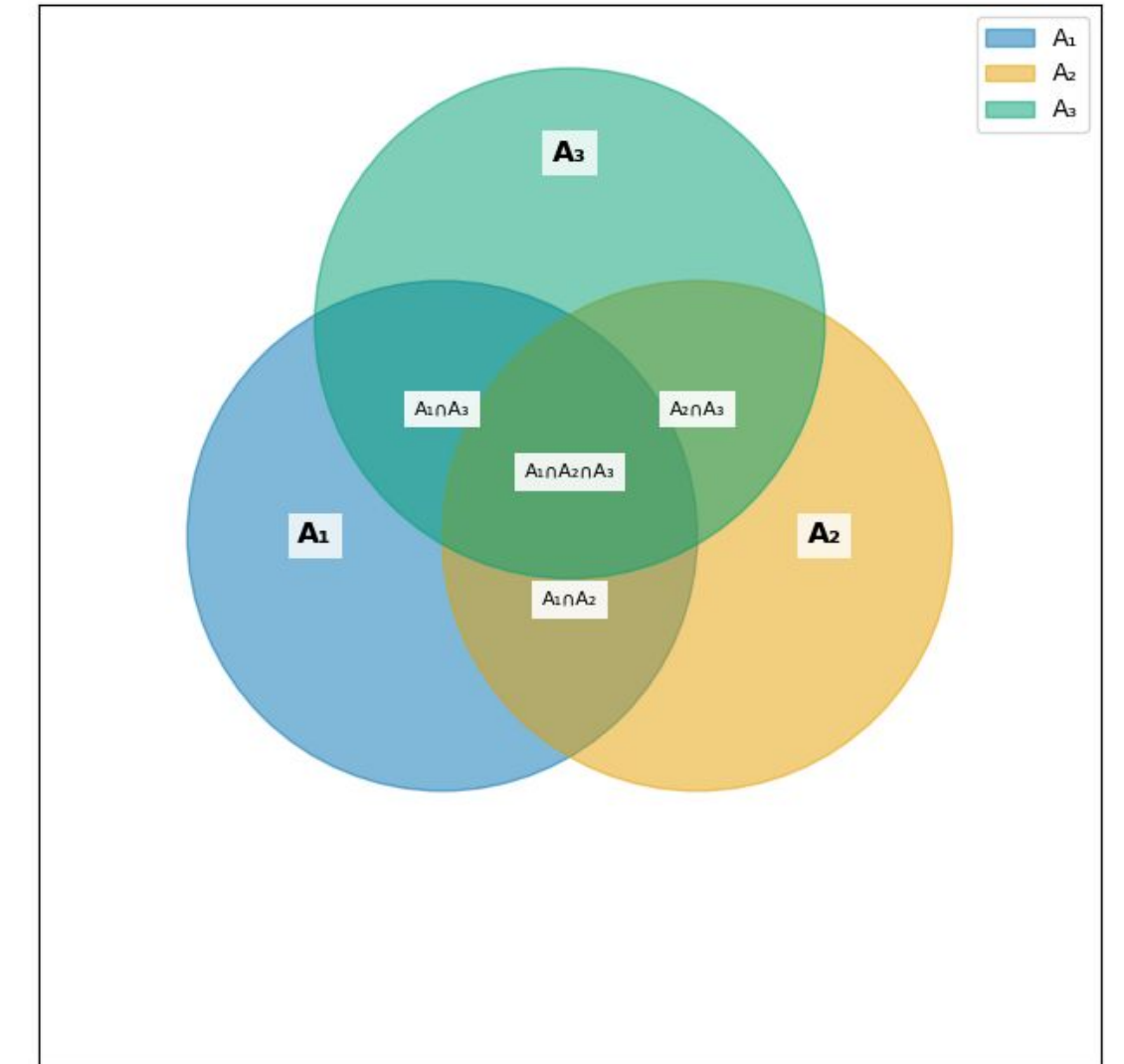
Inclusion exclusion principle says that

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Size of the union is $\leq$ Sum of the sizes of the individual sets

$$|A_1 \cup A_2 \cup A_3| \geq |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| - (2).$$

Size of the union is $\geq$ Sum of the sizes of the individual sets – Sum of the sizes of pairwise intersections



Inclusion Exclusion principle

Suppose we have k finite sets $A_1, A_2, \dots, A_k$

Inclusion exclusion principle says that

$$|A_1 \cup A_2 \cup A_3 \dots \cup A_k| \leq \sum_{\ell=1}^k |A_{\ell}| \quad -(1')$$

$$|A_1 \cup A_2 \cup A_3 \dots \cup A_k| \geq \sum_{\ell=1}^k |A_{\ell}| - \sum_{1 \leq j < \ell \leq k} |A_j \cap A_{\ell}| \quad -(2').$$

Question: What does this have to do with probabilities?

Inclusion Exclusion principle for probability spaces

Suppose we have k events $E_1, \dots, E_k$.

Inclusion exclusion principle says that

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \leq \sum_{\ell=1}^k \Pr[E_\ell] \quad -(1'')$$

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \geq \sum_{\ell=1}^k \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq k} \Pr[E_j \cap E_\ell] \quad -(2'').$$

$E_1 \cup E_2 \cup \dots \cup E_k$: means at least one of the events $E_1, \dots, E_k$ occurs.

$E_j \cap E_\ell$: means both the events E_j and E_ℓ occur.

Inclusion Exclusion principle for probability spaces

Suppose we have k events $E_1, \dots, E_k$.

Inclusion exclusion principle says that

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \leq \sum_{\ell=1}^k \Pr[E_\ell] \quad - (1'')$$

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \geq \sum_{\ell=1}^k \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq k} \Pr[E_j \cap E_\ell] \quad - (2'').$$

Equation (1'') is called the “Union bound” in probability theory – very powerful tool!

Application: Inclusion-Exclusion principle

Recall the application – computing fixed points of a random permutation

We have : $A = \{1, 2, \dots, n\}$ and let $\mathbf{B}$ be a random permutation of A .

We are interested in the number of fixed points of $\mathbf{B}$ (denoted by $\text{FP}(\mathbf{B})$)

– i.e., positions i which are unchanged in $\mathbf{B}$.

Previously: we have shown $E[\text{FP}(\mathbf{B})] = 1$ and $\Pr[\text{FP}(\mathbf{B}) \geq c] \leq \frac{1}{c}$.

Question: What is the probability that $\text{FP}(\mathbf{B}) \geq 1$?

Number of fixed points of a random permutation

We have shown $E[\text{FP}(\mathbf{B})] = 1$ and $\Pr[\text{FP}(\mathbf{B}) \geq c] \leq \frac{1}{c}$.

What is the probability that $\text{FP}(\mathbf{B}) \geq 1$? Is it at least *say* 0.1?

Scenario 1: $\Pr[\text{FP}(\mathbf{B}) = 0] = 1 - \frac{1}{n}$ and $\Pr[\text{FP}(\mathbf{B}) = n] = \frac{1}{n}$?

Note that in scenario #1, $\Pr[\text{FP}(\mathbf{B}) \geq 1] = \frac{1}{n}$.

We will show that **Scenario 1 is not possible!**

Lower bounding the probability of having at least one fixed point

Recall our notation from previous video:

$FP_j(\mathbf{B}) = 1$ if j is a fixed point in $\mathbf{B}$ and 0 otherwise.

Define the event E_j to be when $FP_j(\mathbf{B}) = 1$.

Observation #1: The event $FP(\mathbf{B}) \geq 1$ is the same as the event $E_1 \cup E_2 \dots \cup E_n$.

Applying Inclusion Exclusion

$FP_j(\mathbf{B}) = 1$ if j is a fixed point in $\mathbf{B}$ and 0 otherwise.

Define the event E_j to be when $FP_j(\mathbf{B}) = 1$.

So, by equation (2'') of Inclusion-Exclusion, we have

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_n] \geq \sum_{\ell=1}^n \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq n} \Pr[E_j \cap E_\ell] \quad -(4).$$

Note that $\Pr[E_\ell] = \Pr[FP_\ell(\mathbf{B}) = 1] = \frac{1}{n} \quad -(5)$

Calculating probabilities of pairwise intersection

What about the probabilities of pairwise intersections?

$\Pr[E_j \cap E_\ell]$ = probability that both j and ℓ are fixed points

A random permutation is equally likely to map j and ℓ to any of $n(n - 1)$ possibilities. Thus, the probability that j is mapped to j and ℓ is mapped to ℓ is exactly $\frac{1}{n(n-1)}$.

Thus, $\Pr[E_j \cap E_\ell] = \frac{1}{n(n-1)}$ — (6).

Putting it all together

Recall equation (4) says

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_n] \geq \sum_{\ell=1}^n \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq n} \Pr[E_j \cap E_\ell].$$

From equation (5), $\Pr[E_\ell] = \frac{1}{n}$ and equation (6), $\Pr[E_j \cap E_\ell] = \frac{1}{n(n-1)}$.

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_n] \geq \sum_{\ell=1}^n \frac{1}{n} - \sum_{1 \leq j < \ell \leq n} \frac{1}{n(n-1)} = n \cdot \frac{1}{n} - \binom{n}{2} \cdot \frac{1}{n(n-1)} = 1 - \frac{1}{2} = \frac{1}{2}.$$

Summary

Thus, we can conclude that for a random

permutation $\Pr[\text{FP}(\mathbf{B}) \geq 1] \geq \frac{1}{2}$.

More generally, inclusion exclusion principle
can be a powerful tool to analyze probabilistic
experiments– especially establishing lower bounds
on probabilities